TRAVERSING THE UNKNOWN: WORMHOLES, TIME, AND THE ARCHITECTURE OF THE MULTIVERSE

Ditipriya Bhattacharyya

Henry M Jackson High School

Independent High School Student Researcher

Bothell, Washington, United States

Email: ditipriyab2009@gmail.com

Abstract

This review paper examines the scientific theory and theoretical implications of wormholes, time travel, and the multiverse, in an attempt to consolidate speculative ideas with current advances in physics and quantum computing. Employing the laws of general relativity, string theory, and quantum mechanics, it critically examines the Einstein-Rosen bridge, the implications of entanglement for traversable wormholes, and responds to the philosophical objections to parallel universes. Furthermore, it highlights current experiments emulating the wormhole-like effects using quantum computers, thus demonstrating the dynamic intersection of theoretical physics and artificial intelligence. By examining the work of physicists like Einstein, Susskind, and Tegmark, this paper provides a comprehensive overview for readers and curious students by pushing the boundaries of space, time, and the fabric of existence, speculating that our universe is only one in a complex multiversal arrangement. This paper addresses questions of multiverse and spacetime while exploring wormholes, multiverses, time travel, and theories that remain purely theoretical yet deeply rooted in the physics realm. While none of these theories or claims have been scientifically proven, they continue to shape speculative research and provoke philosophical inquiry.

Keywords: wormholes, multiverses, quantum mechanics, general relativity, artificial intelligence

Introduction

The concept of a tunnel allowing movement across universes, similar to teleportation, has been captured by both the public and scientific imagination.

In moments of stillness, wrapped in our own thoughts, our mind sometimes wanders off. We imagine possibilities and build theories, creating a world of conjectures. One such idea born from both imagination and scientific theory is the wormhole.

A fair question to ask at this point is, "What are wormholes?" In simple terms, a wormhole is a theoretical tunnel that allows an object to pass from one side to the other in a short time. Wormholes were first theorized by the remarkable physicist Albert Einstein in 1916 using the theory of general relativity [1]. His theory proposed that gravity was not the force pulling objects together but rather an effect of curved spacetime around massive objects. This introduced the concept of bending space to form a tunnel structure, today known as a wormhole.

While still speculative, wormholes are not only thought of as shortcuts in space but also as potential means for time travel. They could allow the movement of an object from one point to another in a relatively shorter period of time than usual. According to general relativity, if one end of the wormhole is situated in a different gravitational field, time at that end could behave independently from time at the other end.

This research explores the theoretical foundation and scientific evolution of wormholes as understood today within the frameworks of general relativity and quantum mechanics, and their possible connection to theories of time travel and the multiverse. It also touches on the use of existing technologies, artificial intelligence, in the simulation, modeling, and conceptualization of these complex structures of spacetime.

Methodology

This paper employs a theoretical literature review methodology to examine existing research on multiverse theories, wormholes, and artificial intelligence in physics. Academic sources were selected from scholarly books, peer-reviewed journals, and scientific publications on quantum mechanics, general relativity, quantum computing, and string theory. The paper focuses on combining established theoretical models rather than displaying original experimental data. No experiments or simulations were administered; previously published work was studied and compared to identify limitations, conceptual connections, and future directions in hypothetical physics research.

Results and Discussion

What are wormholes?

A wormhole is commonly defined as a hypothetical structure that is connected by two distant points [2]. The idea of wormholes first arose from Einstein's theory of general relativity, which suggested that space could be curved and connected in weird ways. In the movie "Interstellar," the wormholes are presented as conceptual tunnels that facilitate rapid travel from one point to another, connecting the main character's journey to a new home in another galaxy. This concept astonished many, and while it was fantastical, it is based on the scientific idea of wormholes being a potential shortcut through spacetime. However, their existence and traversability remain purely theoretical. The first type of wormhole ever theorized was the Schwarzschild wormhole, which was initially created to define the internal structure of a black hole [1]. The Schwarzschild wormhole had two parts: the black hole and the white hole. While gravity in the black hole was known for pulling everything in, the gravity in the white hole had the opposite role of expelling everything out. This was thought to complement each other perfectly; however, it resulted in the wormhole collapsing almost instantly, making it purely theoretical and not practical enough to prove.

The idea of wormholes is deeply rooted in Albert Einstein's theory of general relativity, which redefined gravity not as a force but as a bending of space caused by massive objects. However, since this theory primarily applied to large-scale objects like stars and planets, it started breaking down when applied to smaller entities such as electrons. To address this issue, Nathan Rosen and Albert Einstein started working together in 1935, aiming to remove the singularity, the point where gravity would become infinite. For this, they revisited the Schwarzschild wormholes and started studying the singularity believed to exist in the center of the black hole. When they extended the math, they made a remarkable discovery. They realized that if they mirrored the singularity of a blackhole, it created a spacetime tunnel or bridge connecting two regions of spacetime. This structure was known as the Einstein-Rosen Bridge, the earliest formal model of a wormhole. Although groundbreaking, the Einstein-Rosen Bridge was unstable, not allowing anything light or small to travel through, making it a non-traversable wormhole.

How Could Wormholes Work?

Traversable wormholes should be something unfamiliar to you, so allow me to introduce you to the Morris–Thorne model of traversable wormholes. In 1988, two respectable scientists, Kip Thorne and Michael Morris, created a model of a wormhole that allowed human beings or objects to pass through safely. They showcased the fact that in order for the wormholes to be safe enough, the throat of the wormhole required some matter of negative energy that had the potential to violate the null energy present in the wormhole. This stable version of the model laid out the groundwork for the possibility of bridges across spacetime.

Exotic matter, matter of negative energy, was crucial since it would counteract the strong bending of spacetime and support the wormhole geometry, preventing the collapse of the throat. A well-known example of this kind of negative matter is the **Casimir effect** [3]. This quantum phenomenon allowed less energy than the vacuum outside between two slowly spaced plates, which ultimately resulted in a negative energy. However, in recent times, there have been some particular advances, for example, the Yukawa-Casimir wormhole model is a much better version now since it offers the same amount of negative energy, boosting stability while minimizing the use of exotic matter [3]

Nevertheless, creating or even finding usable exotic matter remains a titanic undertaking. Although phenomena such as squeezed vacuum states in quantum optics and Hawking radiation at the edge of black holes provide enticing sources, these exist only under highly controlled or extreme conditions. Certain theoretical ideas, such as the Alcubierre drive, also depend upon negative energy, showing how crucial exotic matter may turn out to spacetime manipulation, even if it remains purely a conceptual endeavor to date. If we ever develop the capability to use negative energy, the potential is transformative: not only for wormholes, but for reworking portions of general relativity, for enabling faster-than-light travel, or even exploring the stuff of time. [4]

Time travel has captivated scientists and the public alike, often depicted in movies and literature. In the cult-classic, Donnie Darko, time travel has been visualized through a translucent tunnel, which was a fictional yet engrossing interpretation of a wormhole, bending spacetime. Though fictional, this movie portrays that wormholes could not only allow travel through space but also through time. However, this theory contains several complications, one of which is the “grandfather paradox”. In response to this, Stephen Hawking introduced the Chronology Protection Conjecture. This conjecture suggested that quantum effects, such as vacuum fluctuations accumulating at the entrance of wormholes, could destroy the existence of a wormhole even before it could create a closed time loop [2]. Henceforth, even though theories have suggested time traveling, science may not allow it.

Since a major problem for time travel is paradoxes and instability, physicist Matt Visser proposed the theory of Roman Rings in 1997 [6]. It was a configuration consisting of several traversable wormholes, arranged in a circular pattern. While no single wormhole allowed for time travel, together they created a closed timelike curve capable of enabling a time loop. Visser suggested that by using enough wormholes, the quantum back reaction that destabilized time could be minimized, thereby keeping the whole system under the chronology violation. Together, these theoretical advances suggest that if wormholes exist and can be stabilized, they may not only act as bridges in space but also challenge the understanding of time itself.

Multiverses

Moving to our next topic: multiverses.

Quantum cosmology and string theory allow the existence of multiple, possibly infinite universes, in other words called multiverses. String Theory suggests that the universe is not made of point-like particles but rather made of tiny 1-dimensional strings that require it to vibrate in 10 -11 dimensions [5,7]. Since we only have the power to observe 4 dimensions (3 space + 1 time), physicists believe the extra dimensions curl up into a phenomenon called the Calabi-Yau manifolds. These manifolds exist in about 10^500 ways and can result in different types of universes or strengths of forces [8]. This enormous range of possibilities is called the String Landscape, and it serves as one of the strongest theoretical bases for the existence of universes.

On the other hand, Quantum cosmology applies the rule of quantum mechanics to the whole of the universe, especially when it first appeared. Quantum cosmologists believe that the universe appeared from a quantum fluctuation, commonly referred to as the ‘big bang’ [8]. In simpler terms, a quantum fluctuation is a random, tiny burst that originated from nothingness and gave rise to space and time itself. These fluctuations are tiny random variations that can even occur in places of vacuum. They are described by a mathematical function known as a wavefunction, which outlines the millions of possibilities or states the quantum system can take. This theory offers many responses to cosmic evolution, including histories and realities.

MIT physicist Max Tegmark proposed the theory that the multiverse might not be just one singular idea, but has 4 different layers of depth and weirdness attached to it [9]. **Level I** refers to regions of space that are not visible to us. These parts of the universe could be unexplored by us, performing slightly differently, without having any new specific physics laws. “Bubble universes”, that emerged from the theory of eternal inflation, wrap his Level II theory of space. He believed that since space has always been expanding, some parts of space might cease to do so, turning into a bubble and resulting in a universe of their own, possibly with different physical constants and laws. Level III is rooted in quantum mechanics and inspired by Hugh Everett’s theory in 1957, where he recommended that the universe splits into parallel branches. He assumed that everything had the possibility to exist in another universe if not our own. Lastly, his Level IV was based on something purely theoretical, revolving around the idea that it was achievable to have every mathematical structure ever constructed to work at some point in space. Tegmark argues that if a mathematical structure is self-consistent enough, it doesn’t describe a universe; it can be one.

The universe seems perfectly tuned for our existence, which raises a question that has puzzled scientists for years: How is this all perfectly created?

If the gravitational constant were slightly stronger or weaker, stars would not have formed the same way. If the cosmological constant were too large, the universe would be expanding at a much faster rate for galaxies to form. This is where the Anthropic Principle comes in. This principle states that the universe must have some property that allows the existence of life. When we connect this to multiverses, the image becomes clearer. If string theory and quantum cosmology suggest the existence of multiverses, with different physics and varied laws, it would not be surprising if we lived in a universe where everything worked in our favor. Physicists like Leonard Susskind and Steven Weinberg argued that this fine-tuning may be explained by the vast landscape of possible universes, with ours having the ability to produce life [10,11]. And as we dig deeper into how wormholes connect multiverses, we start questioning not only the Anthropic Principle, but also our existence and why.

Connecting Multiverses

Now, if multiverses do really exist, a very important question that arises is whether these multiverses can interact and, if so, how. There have been a bunch of theories stating how wormholes could be the answer to that, creating tunnels between universes. Even though wormholes have been theorized to connect two points in spacetime, they can also act as gateways between entirely separate universes. Also known as universe-hopping, physicists Leonard Susskind and Juan Maldacena discussed this in the context of the ER=EPR conjecture, where entangled particles could be connected using wormholes that could even scale to universes [12].

In 2013, Maldecena and Susskind started investigating the similarities between the two well-known theories, ER and EPR. ER or the Einstein-Rosen Bridge posits the existence of physical wormholes, while EPR, formulated by Einstein, Podolsky, and Rosen in 1935, proposed that two certain points in space were connected weirdly, similar to a “Bluetooth connection” also known as quantum entanglement. Although quantum entanglement doesn’t perfectly create a traversable wormhole like the ER bridge, it shares the same math and physics structure [12]. Entanglement is said to happen across universes, suggesting that if quantum entanglement equates to wormholes, then a broader door opens to the fact that wormholes could also be creating a pathway to multiverses.

Another entrancing idea is that black holes could even be connecting other universes [13]. This idea was first proposed by physicist Nikoden Poplawski when he suggested that the immense collapse inside a black hole could create a big bounce, an exchange that had enough power to create another universe. This theory not only suggests that black holes create a gateway to another universe but also that they could mother a new one. According to the Einstein-Rosen model, if wormholes are really black holes, then they will be able to serve as a portal to parallel universes, extending the idea of general relativity and the mind-blowing multiverse theory.

AI and the Future of Wormholes

Now, as the frontiers of physics expand, artificial intelligence is rapidly emerging, aiding universities and researchers in discovering more about the mind-bending ideas in theoretical physics.

Recently, in a groundbreaking research in 2022, students of Caltech, Harvard, Fermilab, and Google used the help of the Sycamore Quantum processor to build a simulation of a traversable wormhole using a handful of qubits [14]. Qubits are the building blocks of quantum computers [15]. Just like how bits work in the real world, using the digits 0 or 1, qubits use them both together, thanks to something called superposition. This enables the creation of such unworldly matters as wormholes or entanglement, because they act like how they would in the quantum world. Through the framework of the Ads/CFT duality, they encoded quantum information and transmitted it through an entangled system, effectively recreating a wormhole in the form of a simulation, making it the first time anyone has ever done so [12]. The Ads/CFT duality is a trick in theoretical physics that basically simulates a 3d model in a 2d hologram. This made it easier to code the entire idea of a wormhole into a usable simulation.

Beyond simulations such as this, AI has also taken up the role of solving calculations that seem extremely complex for the human brain. Physicists are utilizing the ability of AI to scan through massive mathematical landscapes such as the Calabi-Yau manifolds to find similarities in patterns that could result in traversable wormholes [8]. With AI’s capability to solve differential equations, enhance space-related geometries, and run millions of quantum scenarios in a shorter span of time, it is helping scientists to conclude the theory of wormholes at a much faster pace. If AI could predict the perfect conditions to form stable wormholes and detect their data, along with creating better simulations, it could bring quantum gravity and general relativity closer to a union. If wormholes are really the bridges across spacetime or multiverses, AI might be the helping hand we all need.

Conclusion

The concept of wormholes, which was once just an imagination, has now stemmed from billions of scientific theories. Starting from the theory of general relativity to the ER bridge and even to Kip Thorne’s traversable models, each step has only opened more doors to the unknown. Alongside them, theories such as quantum cosmology or string theory have also introduced the existence of multiverses, governed by different, varied laws of physics and structures of math. When these theories are presented together, wormholes not only act as shortcuts through space and time but also as potential gateways between universes.

As our knowledge grows, so do the capabilities of AI. Artificial Intelligence stands at the frontier, accelerating research and developing new and more advanced laws to determine how the universe and wormholes work. AI is becoming a silent partner in humanity's search for the cosmic truth.

So, are wormholes just bridges or tunnels allowing travel through spacetime? Or can they also act as gateways connecting multiverses and acting as a hidden architecture between them? We may not have been provided with answers yet. However, with the help of physics, mathematics, AI, and the never-ending human curiosity, we aren't very far off from revealing the undiscovered truth.

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